

Q1) Short Questions & MCQ

A. A Full binary tree has exactly 9 nodes. How many leaf nodes does it have?

(1 mark)

Ans.

B. What is the maximum number of nodes possible in a binary tree of height 4?

(1 mark)

Ans.

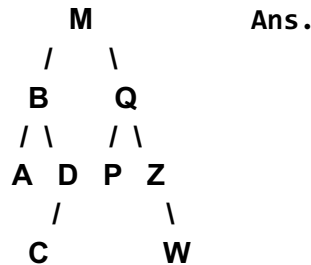
C. If a complete binary tree is represented in an array starting at index 1, what is the index of the parent of the node located at index 13?

(1 mark)

Ans.

D. Write the preorder traversal of this binary tree.

(2 mark)



E. In a complete binary tree represented as an array (where the root node is stored at index 1), what is the formula to find the exact index of the left child of a node located at index i? [Tick the correct option]

(1 mark)

- a) $i+1$
- b) $2*i$
- c) $i / 2$
- d) $2*i + 1$

Q2) Given the roots of two binary trees, root1 and root2, write a function **LevelSumDifference(root1, root2)** to calculate the difference between the sum of all node values at the **odd levels** of the first tree and the sum of all node values at the **even levels** of the second tree. If both the trees are empty return 0. Assume the root node of a tree is at **level 0** (which is considered an even level). The final result should be computed as:

$$\text{Result} = (\text{Sum of odd levels in Tree 1}) - (\text{Sum of even levels in Tree 2})$$

Note: Use recursion. Do not use static or instance variables. You are allowed to use helper functions if needed. A BTNode class is already given. It has 3 instance variables; elem(int type), left (BTNode type), right (BTNode type).

Given Case & Method call	Output	Explanation
Tree 1 ----- <pre> 1 /\ 2 3 /\ 4 5 </pre> Tree 2 ----- <pre> 6 /\ 7 8 \ 9 </pre> LevelSumDifference(root1, root2)	-10	Tree 1 (Odd Levels): The only odd level is Level 1. The nodes at this level are 2 and 3. Sum = 2 + 3 = 5. Tree 2 (Even Levels): The even levels are Level 0 and Level 2. The node at Level 0 is 6. The node at Level 2 is 9. Sum = 6 + 9 = 15 Difference: 5 - 15 = -10

Tree 1 ----- 4 Tree 2 ----- 2 / \ 3 5 / \ \ 7 8 9 LevelSumDifference(root1, root2)	-26	Tree 1 (Odd Levels): There is no odd level. Sum = 0 Tree 2 (Even Levels): The even levels are Level 0 and Level 2. The node at Level 0 is 2. The node at Level 7,8,9. Sum = 2+7+8+9= 26 Difference: 0 - 26 = -26
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